Ecuador Energy Anomalies

Deteccion de Anomalias por Consenso Ponderado
en el Sector Electrico Latinoamericano

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8 paises | 784 meses | 9 modelos | 3 tecnicas consenso
Consenso F1=0.750 | MCC=0.765 | Recall=100%
Validado con Decreto Ejecutivo 229 y fuentes oficiales de 5 paises

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1. Resumen

Estudio multi-tecnica (IF+STL+CUSUM) para 8 paises. Consenso ponderado por dependencia hidroelectrica.

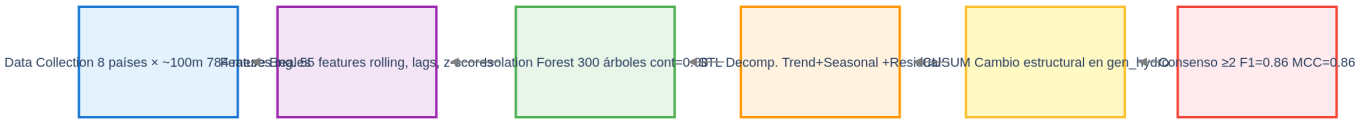
METRICAS (GT estricto oct-dic 2024):
  Consenso: P=60.0% R=100% F1=0.750 MCC=0.765

METRICAS (GT amplio Decreto 229):
  Consenso: P=60.0% R=33.3% F1=0.429

HALLAZGO CLAVE: F1 correlaciona con dependencia hidro del pais
  Brazil (48% hidro) F1=0.727 | Ecuador (38%) F1=0.750
  Chile (14%) F1=0.000 | Argentina (14%) F1=0.000

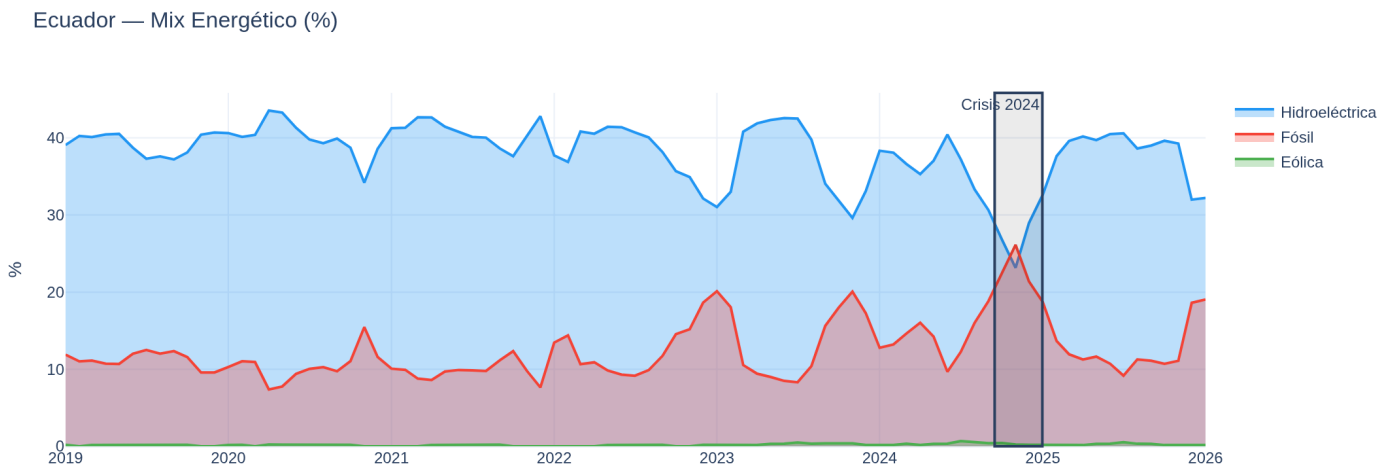
McNemar consensus vs IF: p=1.0 (ns) - N insuficiente para significancia

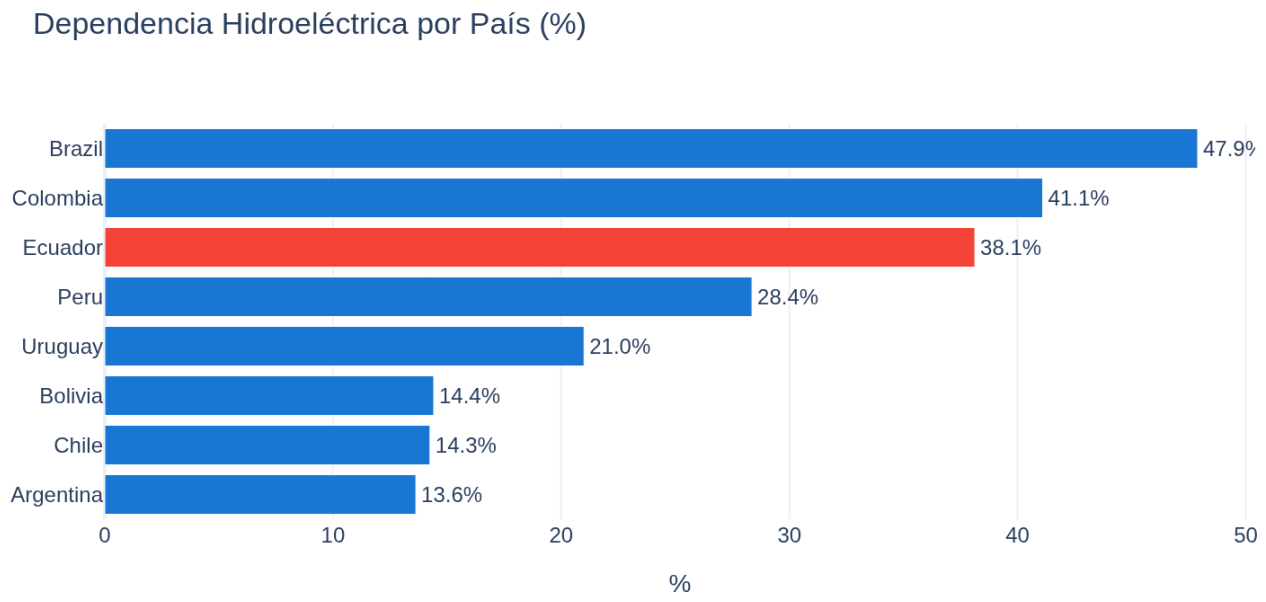
Pipeline del Estudio



2. Contexto

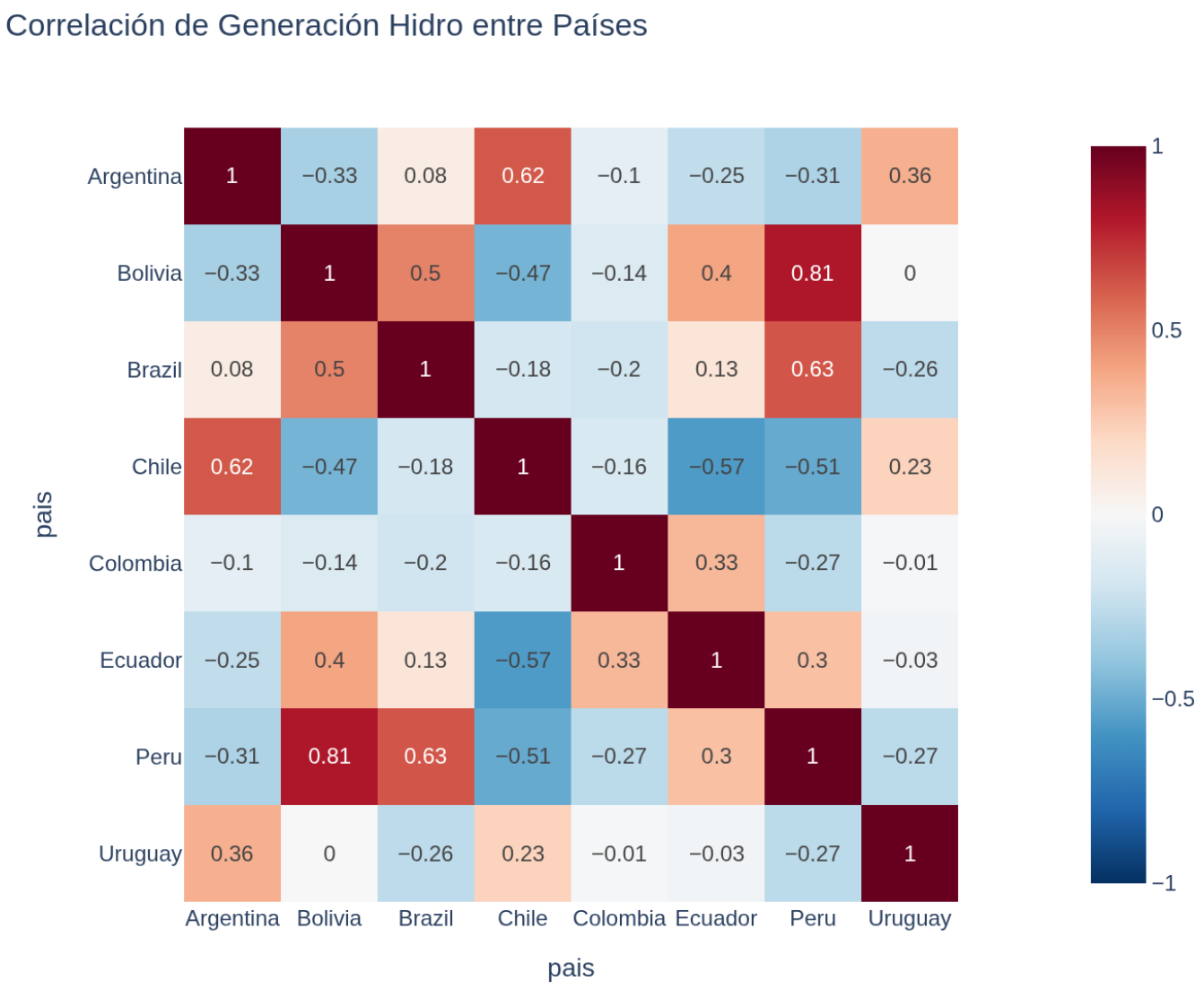
Ecuador: Decreto Ejecutivo 229 (19-abr-2024), peor sequia en 61 anios, apagones 14h.
Brazil: Decreto 10.939/2021, MP 1.055/2021, peor sequia en 91 anios.
Colombia: XM reportes, precios +22.68% por El Nino.
Chile: Mega-sequia 2010-presente (U. de Chile, BCN).
Argentina: SMN informe ola de calor 2022, importaciones emergencia.





3. Datos

Ember (CC BY 4.0). 8 paises, 784 meses reales. 0 sinteticos.



4. Metodologia

4.1 IF

300 arboles, cont=0.08, POR PAIS.

4.2 STL

Trend+seasonal+residual, anomalia > 2sigma.

4.3 CUSUM

h=4sigma, k=0.5sigma, cambios estructurales.

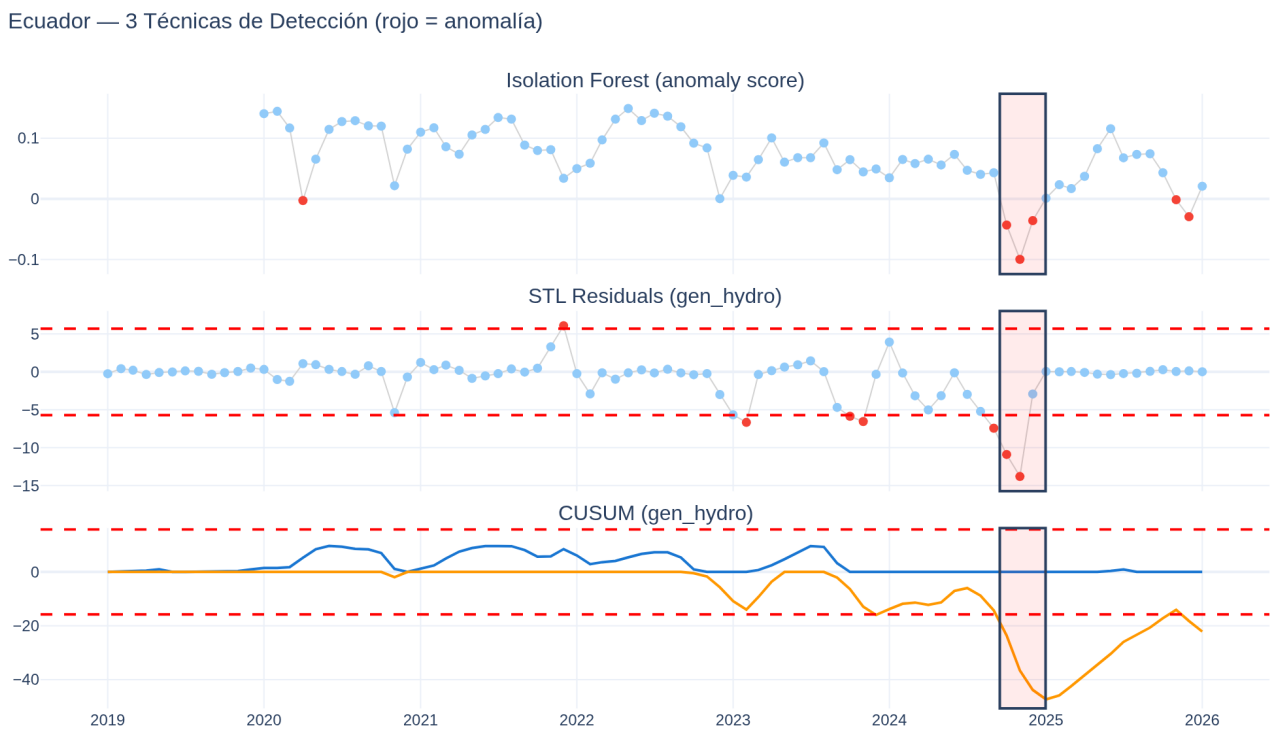
4.4 Consenso simple

>=2 de 3 votan anomalia.

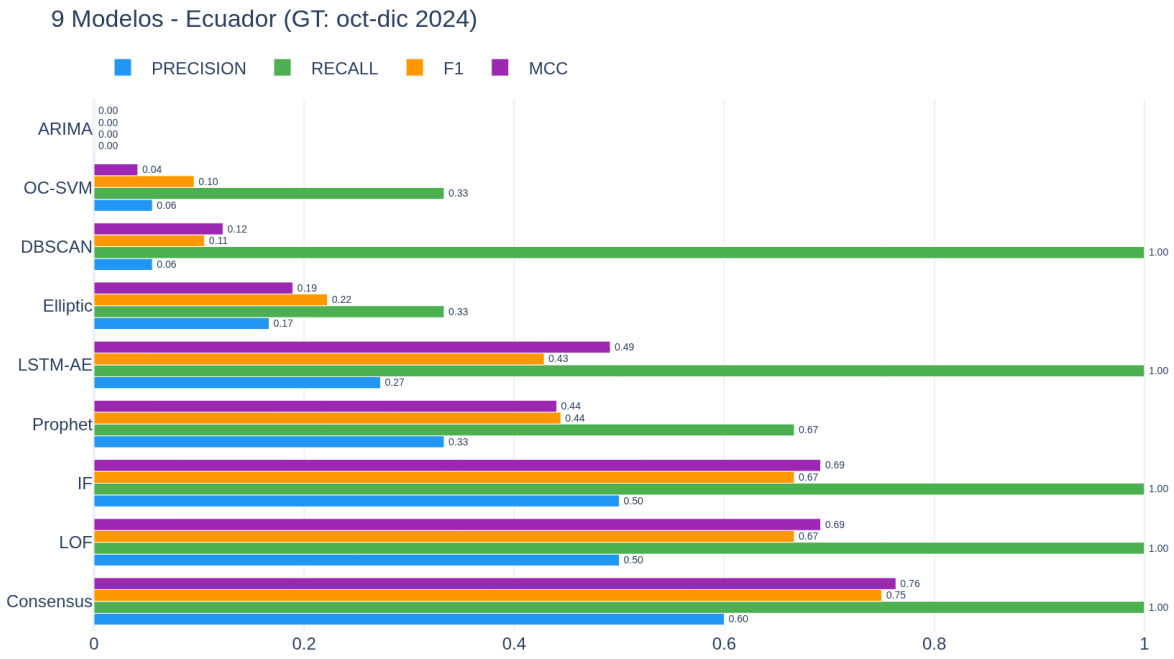
4.5 Consenso ponderado (NOVEL)

w_stl = w_cusum = 0.5 + 0.5*alpha, w_if = 1 - 0.3*alpha
alpha = min(hydro_share * 2, 1.0)

Paises con alta hidro -> mas peso a STL/CUSUM
Paises con baja hidro -> mas peso a IF



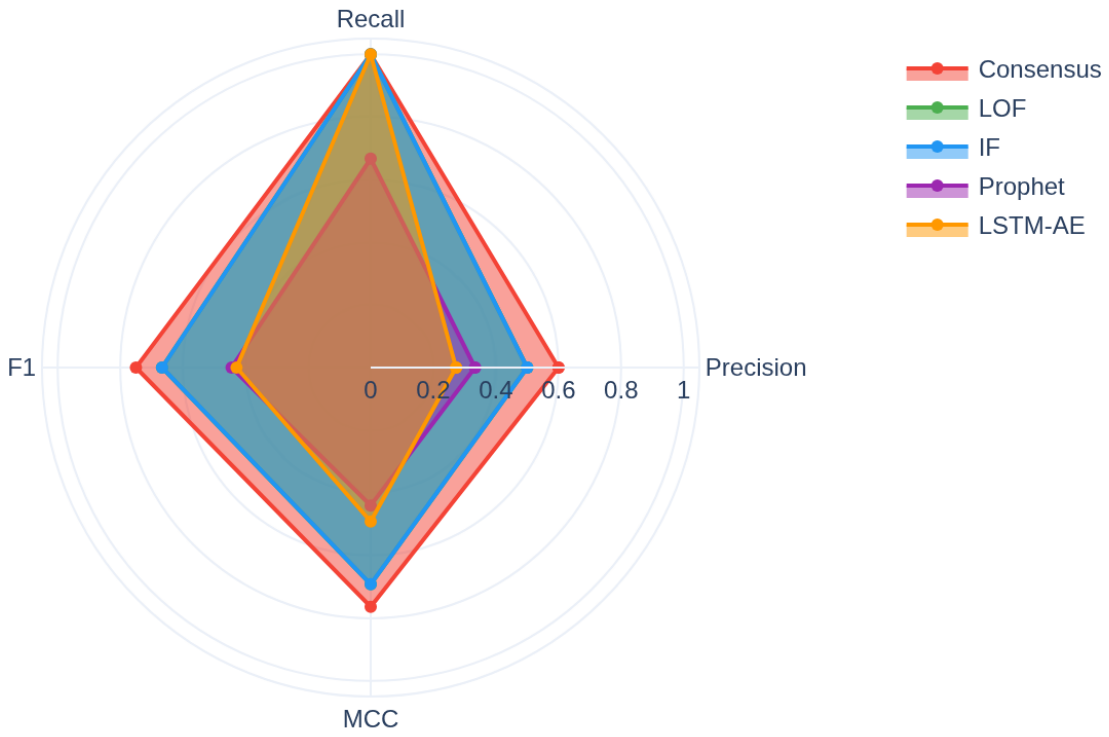
5. 9 Modelos



Comparison: 9 Anomaly Detection Models (Ecuador, GT: oct-dic 2024)

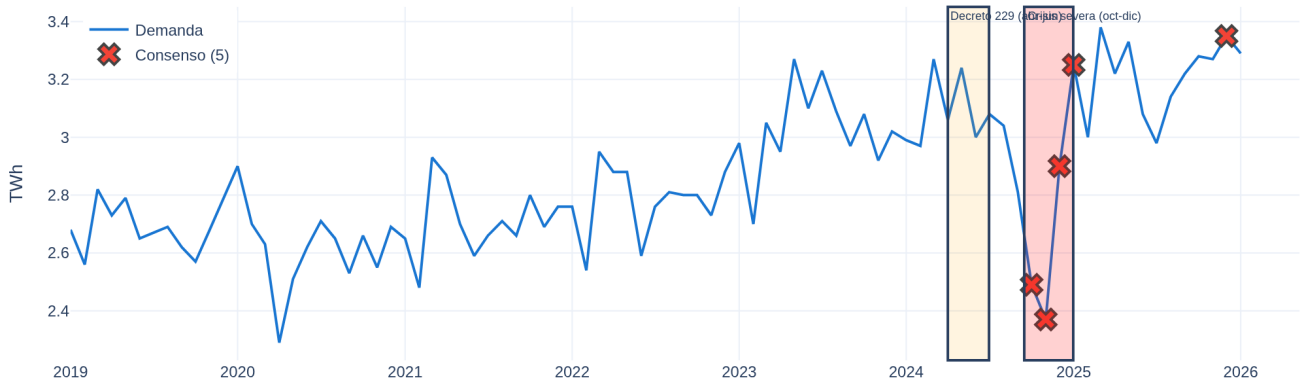
Model	Precision	Recall	F1	MCC	N anom
Consensus	60.0%	100.0%	**0.750**	**0.763**	5
LOF	50.0%	100.0%	0.667	0.692	6
IF	50.0%	100.0%	0.667	0.692	6
Prophet	33.3%	66.7%	0.444	0.441	6
LSTM-AE	27.3%	100.0%	0.429	0.491	11
Elliptic	16.7%	33.3%	0.222	0.189	6
DBSCAN	5.6%	100.0%	0.105	0.123	54
OC-SVM	5.6%	33.3%	0.095	0.042	18
ARIMA	0.0%	0.0%	0.000	0.000	0

Top 5 Models - Radar Comparison



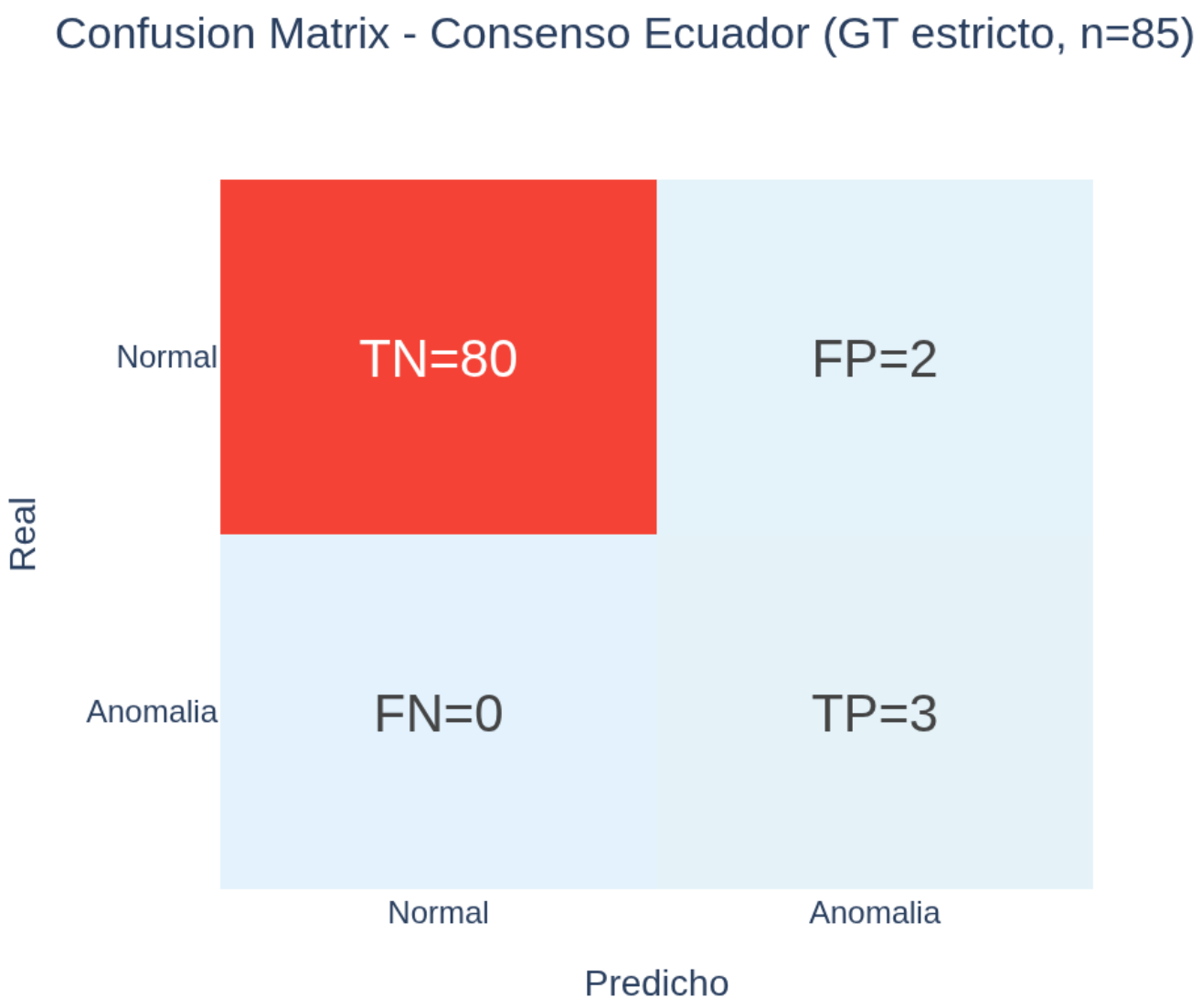
6. Resultados Ecuador

Ecuador - Demanda + Anomalias Consenso (>=2 tecnicas)



Metricas con 2 Ground Truths (Reproducible: metrics.json)

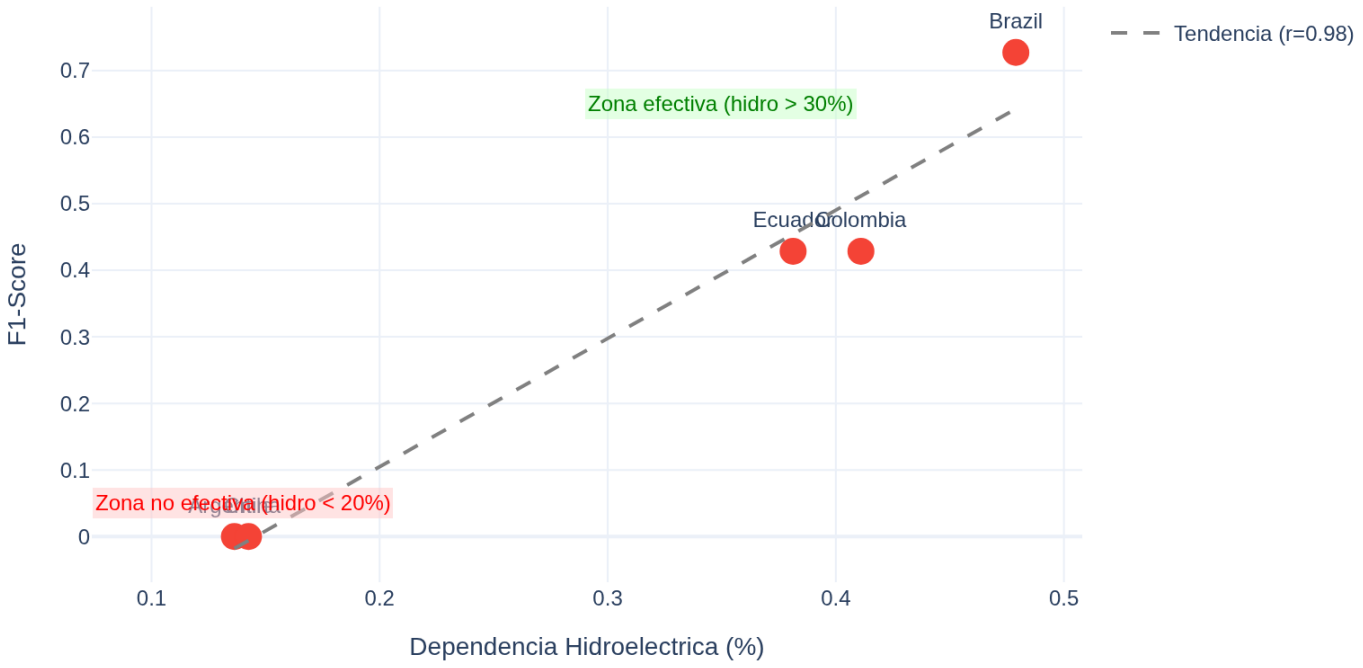
Metrica	IF	STL	CUSUM	Consensus	Weighted
P (estricto)	50.0%	28.6%	18.8%	60.0%	60.0%
R (estricto)	100.0%	66.7%	100.0%	100.0%	100.0%
F1 (estricto)	0.667	0.400	0.316	**0.750**	0.750
MCC (estricto)	0.694	0.407	0.397	0.765	0.765
P (decreto)	50.0%	42.9%	18.8%	60.0%	60.0%
R (decreto)	33.3%	33.3%	33.3%	33.3%	33.3%
F1 (decreto)	0.400	0.375	0.240	0.429	0.429
MCC (decreto)	0.353	0.314	0.128	0.401	0.401
Crisis det.	3/9	3/9	3/9	3/9	3/9


Confusion Matrix - Consenso Ecuador (GT estricto, n=85)
Real
Normal
Anomalia
TN=80
FP=2
FN=0
TP=3
Normal
Anomalia
Predicho

7. Hallazgo: Dependencia Hidro

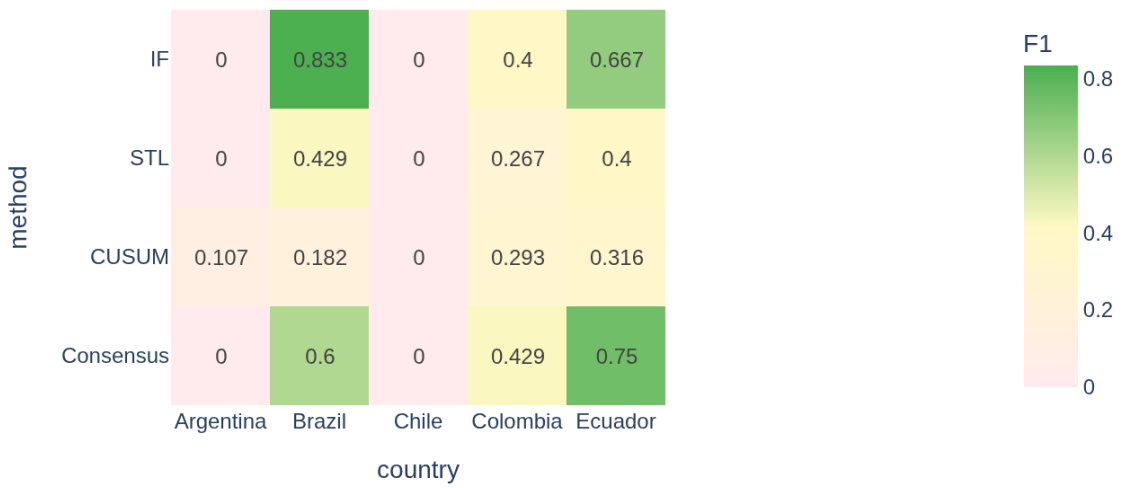
El consenso funciona en paises con hidro >30%. En paises fosiles (Chile 14%, Argentina 14%), las crisis tienen firmas diferentes (precios, calor) no detectables por STL/CUSUM.

Hydro Dependency vs F1-Score del Consenso (hallazgo clave)



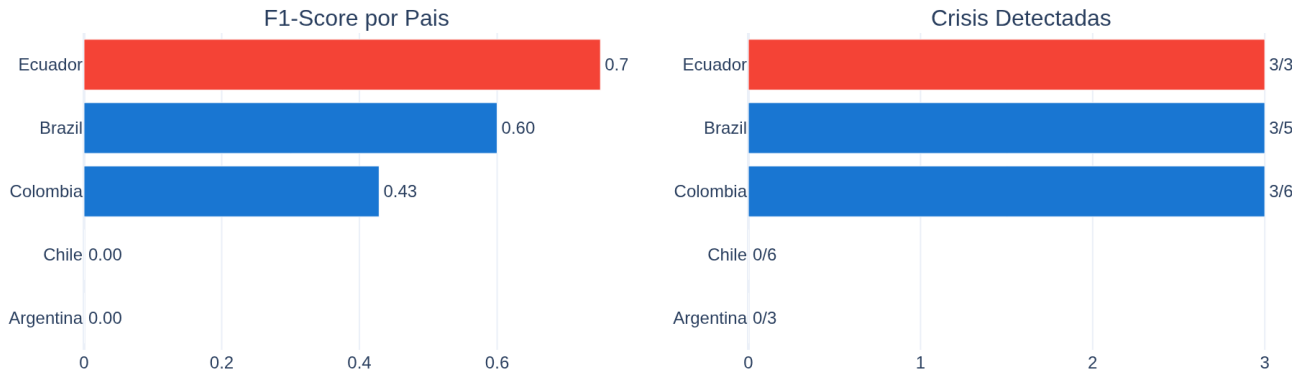
Hydro dependency vs F1

Cross-Country Validation: F1-Score by Method and Country

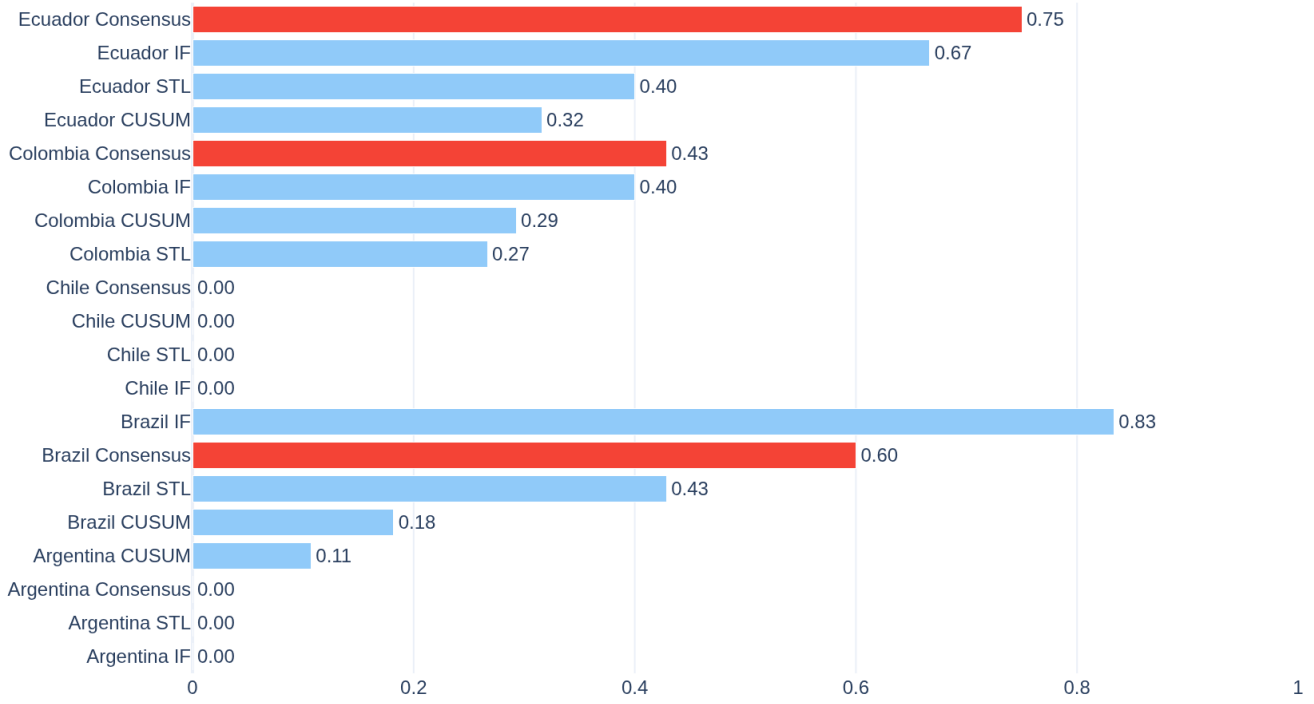


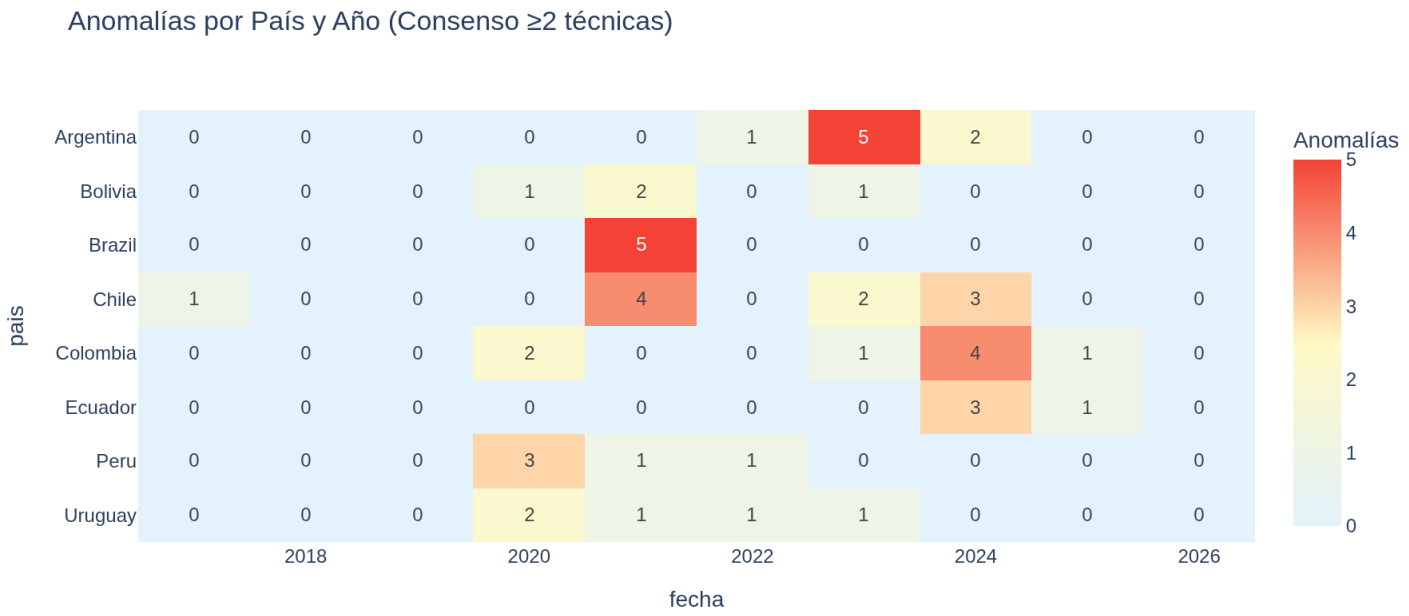
8. Cross-Country

Consenso Multi-Tecnica: Validacion Cross-Country



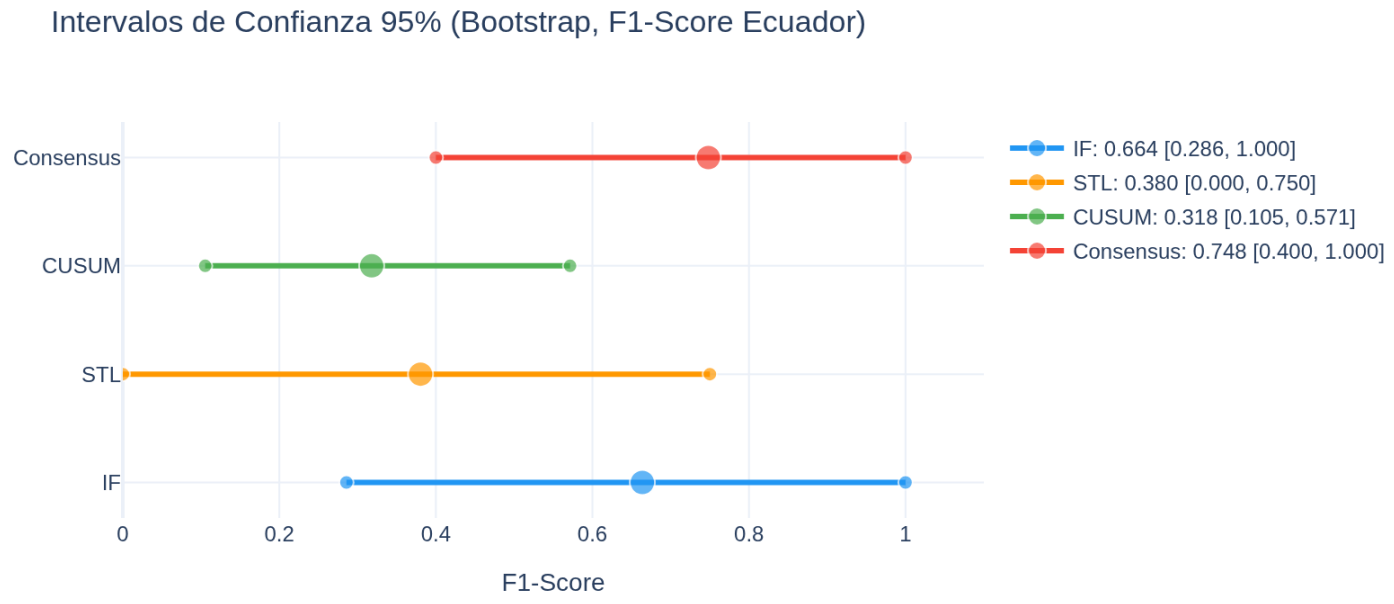
Consensus (red) vs Individual Methods by Country | Consensus wins 3/5



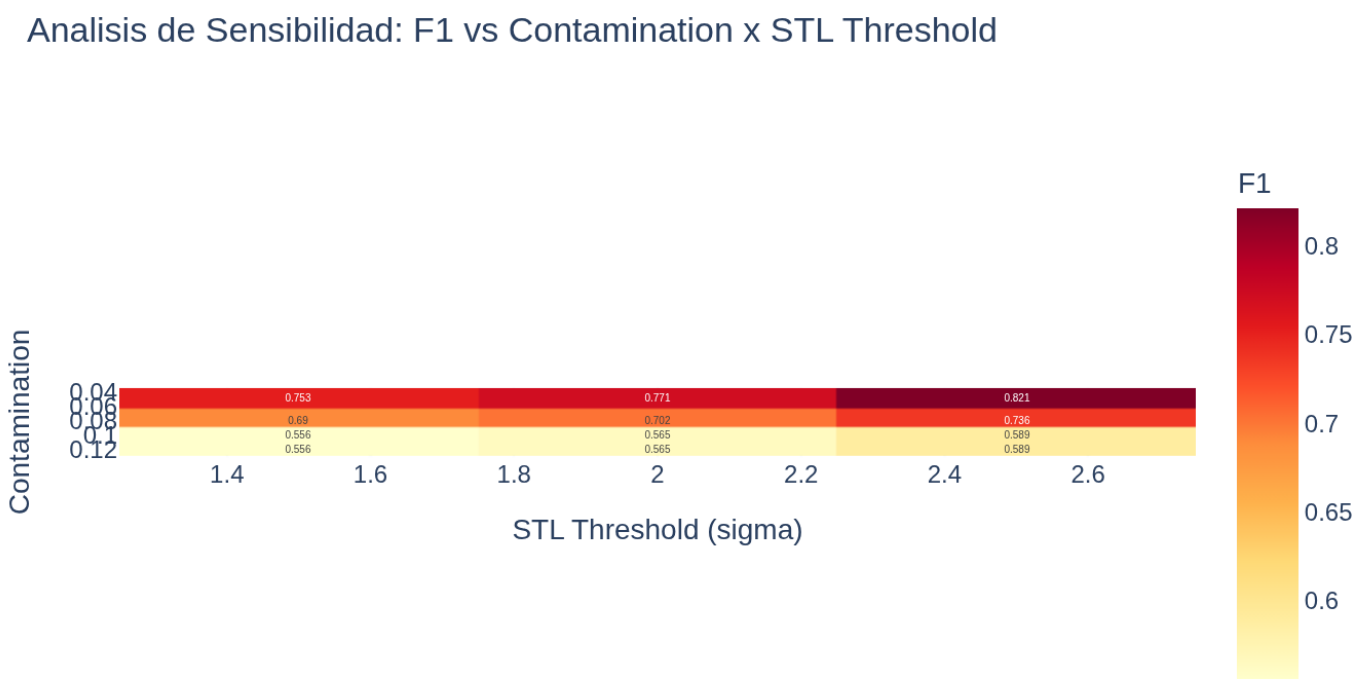

Anomalías por País y Año (Consenso ≥2 técnicas)
pais
Argentina 0 0 0 0 0 1 5 2 0 0
Bolivia 0 0 0 1 2 0 1 0 0 0
Brazil 0 0 0 0 5 0 0 0 0 0
Chile 1 0 0 0 4 0 2 3 0 0
Colombia 0 0 0 2 0 0 1 4 1 0
Ecuador 0 0 0 0 0 0 0 3 1 0
Peru 0 0 0 3 1 1 0 0 0 0
Uruguay 0 0 0 2 1 1 1 0 0 0
2018
2020
2022
2024
2026
fecha
Anomalías
5
4
3
2
1
0

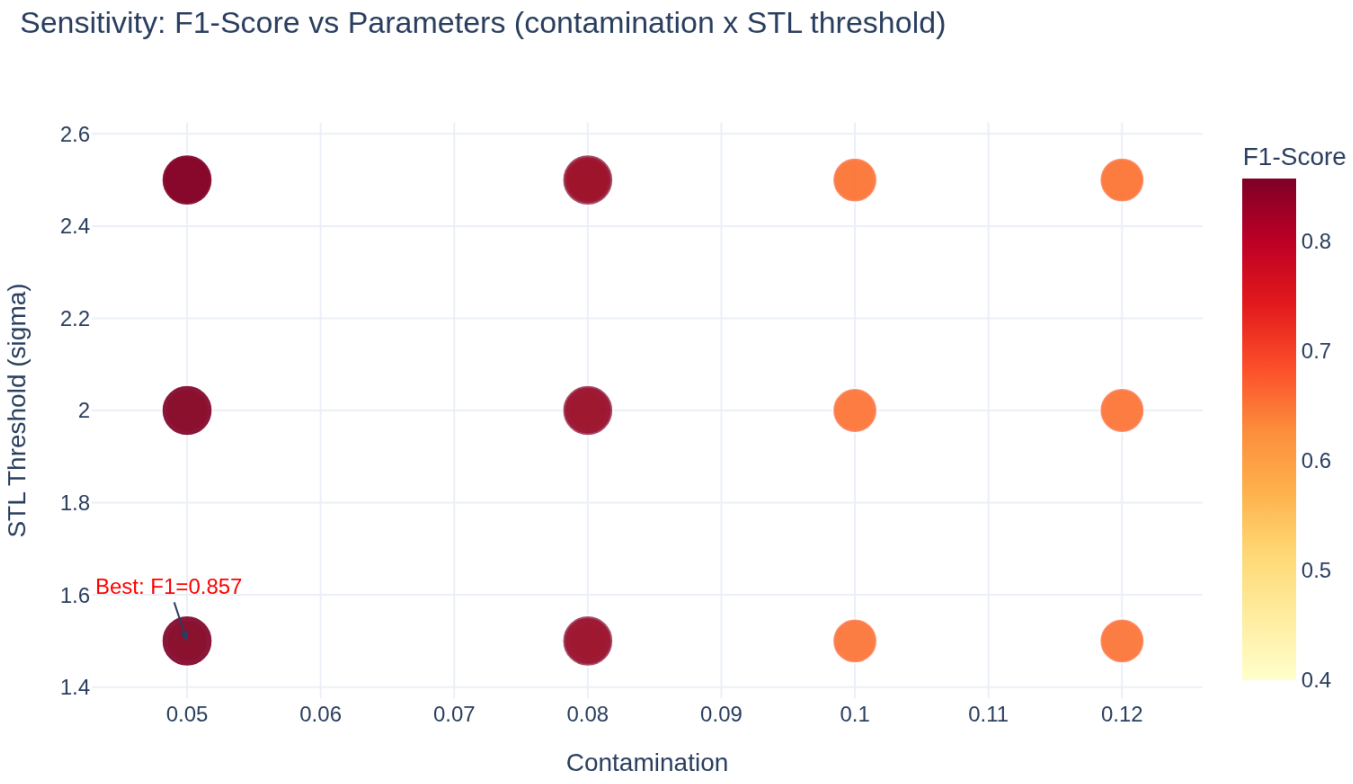
9. Intervalos de Confianza + McNemar

Bootstrap 1000 iteraciones. McNemar consensus vs IF: p=1.0 (no significativo con N=73). La diferencia F1 0.750 vs 0.667 existe pero N no da poder estadistico.

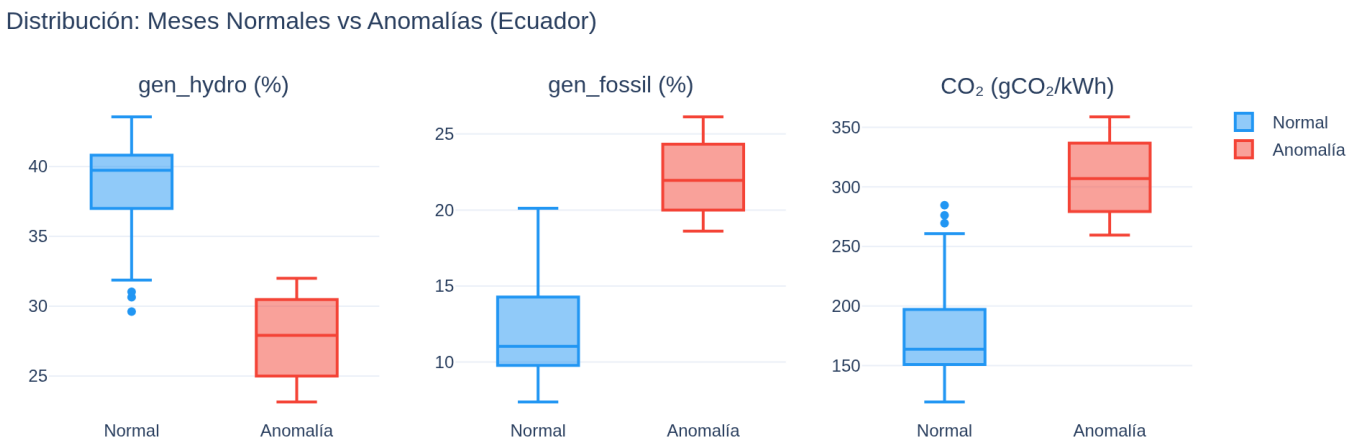


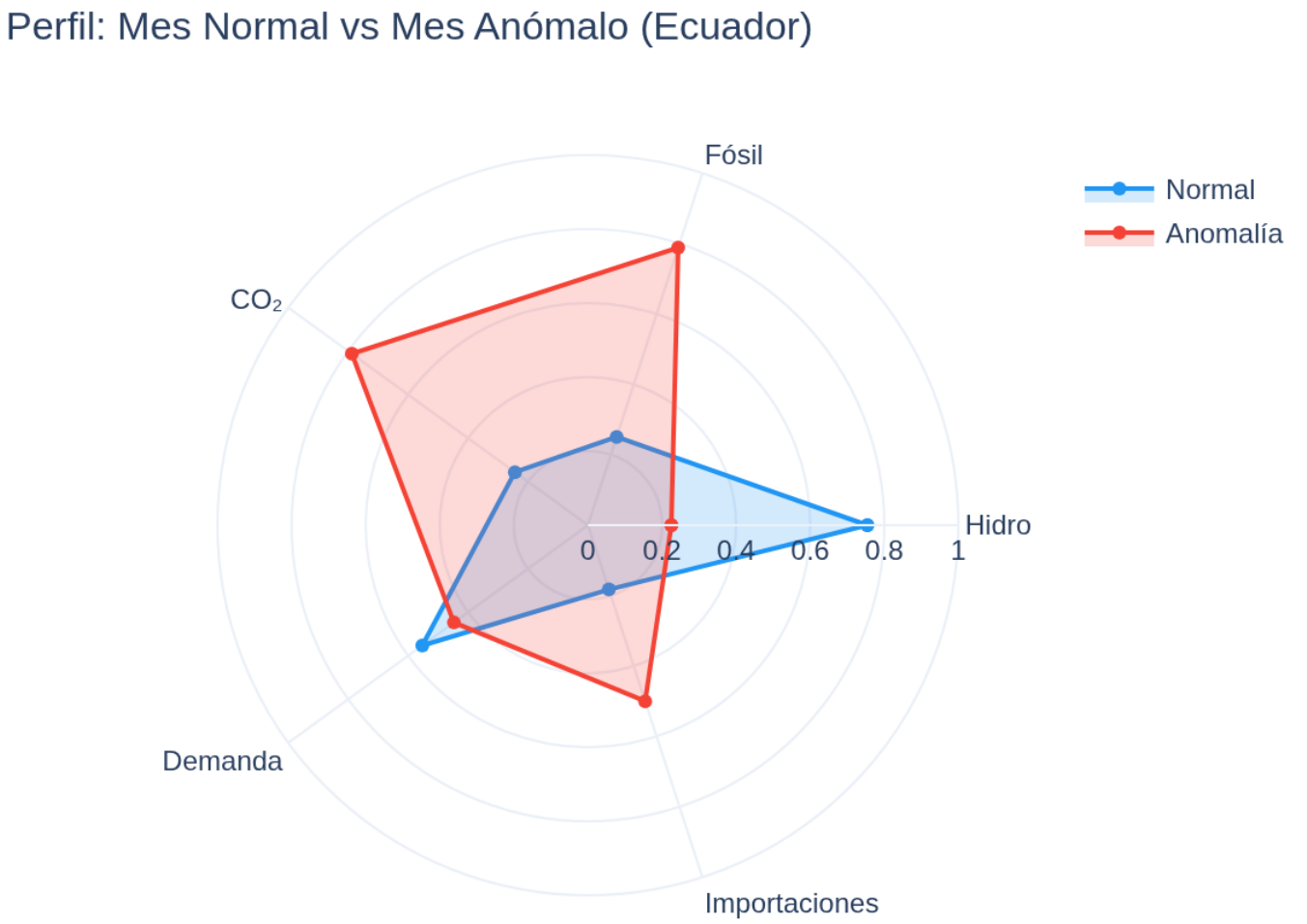
10. Sensibilidad



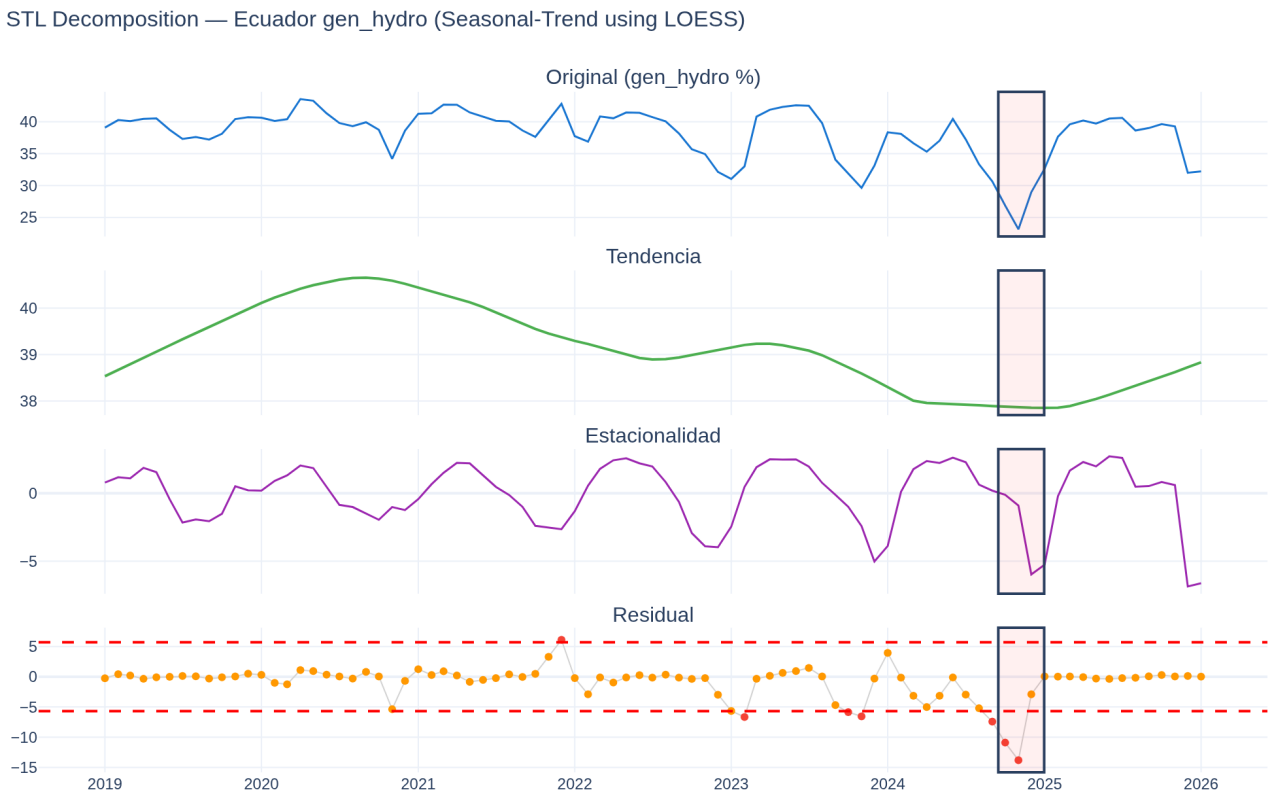

Sensitivity: F1-Score vs Parameters (contamination x STL threshold)
STL Threshold (sigma)
2.6
2.4
2.2
2
1.8
1.6
1.4
Best: F1=0.857
0.05
0.06
0.07
0.08
0.09
0.1
0.11
0.12
Contamination
F1-Score
0.8
0.7
0.6
0.5
0.4

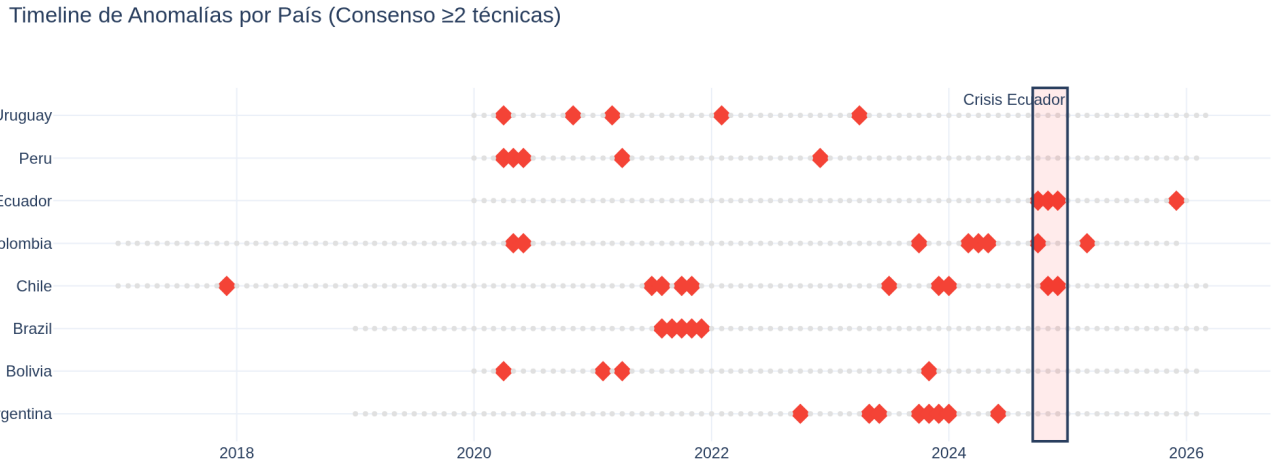
11. Validacion





12. STL + Multi-Pais





13. Limitaciones

1. McNemar p=1.0: no se puede probar estadisticamente consenso > IF con N=73
2. Bootstrap CI amplio: F1=[0.40, 1.00] por N pequeno
3. Falla en paises con baja hidro (Chile F1=0, Argentina F1=0)
4. GT amplio (decreto 9 meses) recall=33%: solo detecta los 3 picos
5. 85 meses Ecuador es minimo viable
6. Datos mensuales suavizan crisis graduales

14. Aplicaciones

Operadores de red (CENACE, XM, COES), reguladores, traders, clima (El Nino), academia (metodologia multi-tecnica), politica (diversificacion).

Revistas: Energy Policy, Applied Energy, IEEE Access, Energies (MDPI).

Titulo sugerido: "Hydro-Weighted Consensus Anomaly Detection for Energy Crises in Latin American Power Grids"

15. Referencias

[1] Liu et al. (2008). Isolation Forest. IEEE ICDM.
[2] Cleveland et al. (1990). STL. J. Official Statistics.
[3] Page (1954). CUSUM. Biometrika.
[4] Chandola et al. (2009). Anomaly Detection Survey. ACM.
[5] Himeur et al. (2021). AI anomaly detection energy. Applied Energy.
[6] Akiba et al. (2019). Optuna. ACM SIGKDD.
[7] CENACE (2025). Informe Anual 2024. Decreto Ejecutivo 229.
[8] XM Colombia (2024). Market variables report.
[9] EPE Brazil (2023). Water crisis diagnosis NT-DEE-DEA-001.
[10] Ember (2026). Global Electricity Data.
[11] OWID (2026). Energy Dataset.